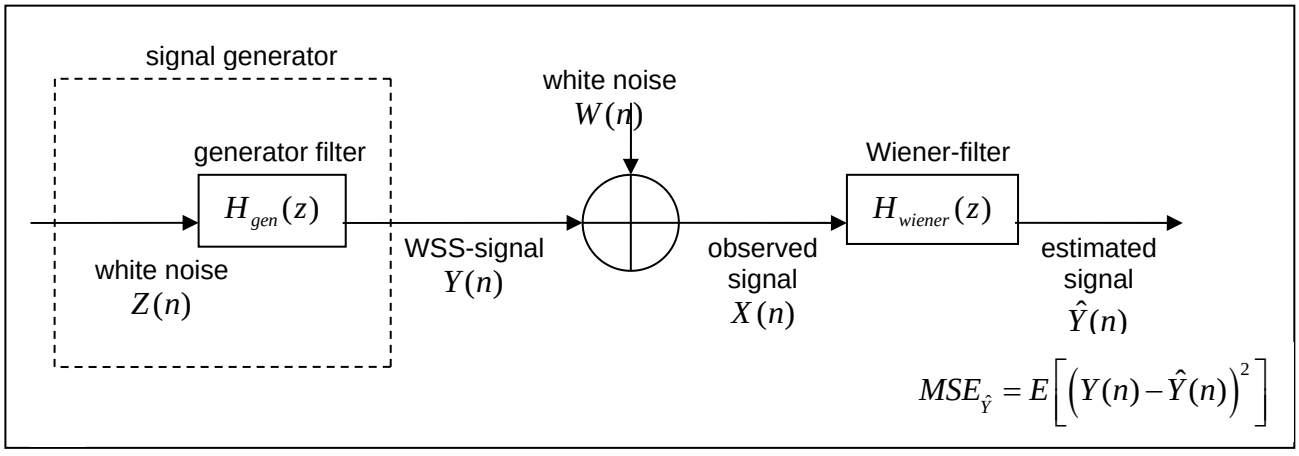
Problem 4.1 (Theoretical + MATLAB)
Signal estimation with discrete-time Wiener-filter (FIR/IIR)

Motivation

This exercise goes through all the most important fundamental issues regarding Wiener-filters, i.e. stochastic optimal estimation (LMMSEE) of noise infected WSS-signals, including the performance of the WSS-filters – both theoretically and as with simulation in MATLAB.
In consideration of comparisons between theoretical calculations and simulations the discrete-time case is treated, but all results are absolutely parallel for the continuous-time case.
The Wiener-filter can, besides direct application for noise reduction, be used as reference when comparing the performance for non-optimal noise reductions methods.

Signal model for the exercise

Below is shown a block diagram for generating a WSS signal, noise infected observation of this, and Wiener-filtering of the observed signal with regard to estimation of the WSS signal.



Note that the signal generator part of the figure only is included due to technicality of the exercise in order to generate a WSS signal $Y(n)$.
In practical applications only the observed signal will be shown $X(n)$ and known or estimated knowledge of the stochastic characteristics for the signal $Y(n)$ and the additive noise $W(n)$.

Structure of the exercise

The exercise has the following sequences:

A. Generation and analysis of the WSS-signal $Y(n)$
B. Generation and analysis of the WSS-noise $W(n)$
C. Generation and analysis of the observed signal $X(n)$
D. **Causal FIR** Wiener filter: design, filtering, analysis, performance
E. **Non-causal FIR** Wiener filter: design, filtering, analysis, performance
F. **Non-causal IIR** Wiener filter: design, filtering, analysis, performance
G. **Causal IIR** Wiener filter: design, filtering, analysis, performance
H. Conclusion concerning performance comparisons.

Both theoretical calculations and calculations with MATLAB is to be performed for all parts.

A. Generation and analysis of the WSS-signal $Y(n)$

The stochastic WSS signal $Y(n)$ is given by

$$R_{yy}(k) = h^{|k|}, \quad h = 0.8 \qquad \text{(autocorrelation)}$$

$$S_{yy}(z) = Z\left\{R_{yy}(k)\right\} = \frac{1-h^2}{(1+h^2)-h(z+z^{-1})} = \frac{1-h^2}{(1-hz^{-1})(1-hz)} \qquad \text{(PSD in z-domain)}$$

$$S_{yy}(\omega) = S_{yy}(z)\Big|_{z=e^{j\omega T_s}} = \frac{1-h^2}{(1+h^2)-2h\cos\omega T_s}, \quad f_s = 1/T_s = 10kHz \qquad \text{(PSD in the frequency domain)}$$

1) Outline $R_{yy}(k)$ and $S_{yy}(\omega)$

2) Demonstrate that $Y(n)$ can be produced by means of the differential equation
$Y(n) = h \cdot Y(n-1) + Z(n)$
where $Z(n)$ (”driving noise”) is white, additive, zero-mean gaussic noise
(help: compute $R_{yy}(k)$ based on the differential equation)

3) Find the required noise effect of driving noise $\sigma_Z{}^2 = E\left[Z^2(n)\right]$ which ensures that $R_{yy}(0) = 1$ as required in the specification for $Y(n)$ above

4) State the transfer function $H_{gen}(z) = \frac{Y(z)}{Z(z)}$ for the signal generator filter

5) MATLAB-realisation:
   - Generate N=1000 samples of driving noise $Z(n)$ (remember to use the calculated $\sigma_Z{}^2$)
   - Filtrate $Z(n)$ with signal generator signal $H_{gen}(z)$
   - Plot the resulting WSS signal $Y(n)$ and its autocorrelation function and PSD
   - Use, e.g., the commands ***randn, filter, plot/stem, xcorr('biased'), freqz***
   - Compare with the theoretical results and comment.

B. Generation and analysis of the WSS-noise $W(n)$

The additive, white, zero-mean, gaussic noise $W(n)$ is given by

$$W(n) = N(0, \sigma_W{}^2), \quad \sigma_W{}^2 = 4$$

$$S_{ww}(z) = \sigma_W{}^2$$

Furthermore, the noise $W(n)$ is uncorrelated with the signal $Y(n)$ ($R_{yw}(k) = 0, \quad \forall k \in Z$)

6) Outline $R_{ww}(k)$ and $S_{ww}(\omega)$

7) MATLAB-realisation:
- Generate N=1000 samples of the noise $W(n)$ (remember to apply $\sigma_W{}^2$)
- Plot the noise $W(n)$ and its autocorrelation function and PSD
- Use, e.g., the commands ***randn, plot/stem, xcorr('biased'), freqz***
- Compare with the theoretical results and comment.

C. Generation and analysis of the observed signal $X(n)$

The observed (noise infected) signal $X(n)$ is given by

$$X(n) = Y(n) + W(n)$$

8) Demonstrate, that under the given assumptions, the following apply

$$R_{xx}(k) = R_{yy}(k) + R_{ww}(k) \quad \text{and thereby}$$

$$S_{xx}(z) = S_{yy}(z) + S_{ww}(z) \quad \Rightarrow \quad S_{xx}(\omega) = S_{yy}(\omega) + S_{ww}(\omega)$$

9) Demonstrate, that under the given assumptions, the following apply

$$R_{xy}(k) = R_{yy}(k) \quad \text{and thereby}$$

$$S_{xy}(z) = S_{yy}(z) \quad \Rightarrow \quad S_{xy}(\omega) = S_{yy}(\omega)$$

10) Demonstrate the estimation error by means of using $X(n)$ for the estimation of $Y(n)$, i.e. state $MSE_X = E\left[\left(Y(n) - X(n)\right)^2\right]$

11) MATLAB-realisation:
- Generate the observed signal $X(n)$
- Plot $X(n)$ and its autocorrelation function and PSD
- Calculate the estimation error MSE_X
- Use, e.g., the commands ***mean, plot/stem, xcorr('biased'), freqz***
- Compare with the theoretical results and comment.

D. Causal FIR Wiener filter: design, filtering, analysis, performance

Now, the signal $Y(n)$ is to be estimated based on the observed signal $X(n)$ at filtration of $X(n)$ with a causal FIR Wiener-filter, defined by

$$H_{wiener}(z) = \sum_{m=0}^{M} h(m) \cdot z^{-m}$$

As shown in the figure on page 1, the Wiener-filtered signal $\hat{Y}(n)$ as an optimal estimate of $Y(n)$.

For the manual calculations $M = 1$, i.e. the sought FIR-filter has two coefficients $h(0)$ and $h(1)$.

12) Set up the Wiener-Hopf equations and find the optimal filter coefficients $h(0)$ and $h(1)$

13) Calculate the expected estimation error by means of using $\hat{Y}(n)$ for estimation of $Y(n)$ when using the calculated optimal FIR-filter, i.e. calculate

$$MSE_{\hat{Y}} = E\left[\left(Y(n) - \hat{Y}(n)\right)^2\right]$$

Compare the value with $MSE_X = E\left[\left(Y(n) - X(n)\right)^2\right]$

14) State the principle for (no need for making calculations) how a signal-noise relationship could be calculated by using $X(n)$ and $\hat{Y}(n)$ respectively, as estimator for $Y(n)$, i.e. $SNR_{in} = \frac{P_Y}{P_W}$ and

$SNR_{out} = \frac{P_{\hat{Y}}}{P_{W_{out}}}$ where $P_{W_{out}}$ is the effect of the filtered noise signal

15) MATLAB-realisation ($M = 1$):

- Generate $\hat{Y}(n)$ at filtering of $X(n)$ with the calculated FIR-filter
- Plot $\hat{Y}(n)$ and its autocorrelation function and PSD
- Calculate the estimation error $MSE_{\hat{Y}}$ from the concrete signals $Y(n)$ and $\hat{Y}(n)$
- Use, e.g., the commands ***filter, mean, plot/stem, xcorr('biased'), freqz***
- Compare with the theoretical results and comment.

16) MATLAB-calculation ($M > 1$):

- Set up the general incidence $M > 1$ Wiener-Hopf equations for determination of the filter coefficients $\{h(m), \quad m = 0,1,...,M-1\}$ for the actual signal model.
- Implement this in MATLAB so that the filter coefficients can be calculated for a random value of M
- Calculate the theoretically expected value of $MSE_{\hat{Y}}$ for $M = 1,2,...,10$ and plot this as a function of M
- What does the situation $M \to \infty$ correspond to?

E. Non-causal FIR Wiener filter: design, filtering, analysis, performance

Now the signal $Y(n)$ is to be estimated from the observed signal $X(n)$ at filtering of $X(n)$ with a non-causal FIR Wiener-filter defined by

$$H_{wiener}(z) = \sum_{m=-M}^{M} h(m) \cdot z^{-m}$$

As demonstrated in the figure on page 1, the Wiener-filtered signal $\hat{Y}(n)$ is applied as an optimal estimate of $Y(n)$.

For the manual calculations $M = 1$ is used, i.e. the sought FIR-filter has three coefficients $h(-1)$, $h(0)$ and $h(1)$.

17) Set up the Wiener-Hopf equations and find the optimal filter coefficients $h(-1), h(0), h(1)$.

Does the FIR-filter have a linear phase? Why (not) ?

18) Calculate the expected estimation error by means of using $\hat{Y}(n)$ for estimation of $Y(n)$ when using the calculated optimal FIR-filter, i.e. calculate

$$MSE_{\hat{Y}} = E\left[\left(Y(n) - \hat{Y}(n)\right)^2\right]$$

Compare the value with $MSE_X = E\left[\left(Y(n) - X(n)\right)^2\right]$

19) MATLAB-realisation ($M = 1$):

- Generate $\hat{Y}(n)$ when filtering $X(n)$ with the calculated FIR-filter (<u>remember that the filtering shall be performed non-causally!!!</u>)
- Plot $\hat{Y}(n)$ and its autocorrelation function and PSD
- Calculate the estimation error $MSE_{\hat{Y}}$ from the concrete signals $Y(n)$ and $\hat{Y}(n)$
- Use, i.e., the commands ***filter, mean, plot/stem, xcorr('biased'), freqz***
- Compare the theoretical results and comment.

20) MATLAB-calculation ($M > 1$):

- Set up the general case $M > 1$ Wiener-Hopf equations for determination of the filter coefficients $\{h(m), \quad m = -M, ..., -1, 0, 1, ..., M\}$ for the actual signal model
- Implement this in MATLAB so that the filter coefficients can be calculated for a random value of M
- Calculate the theoretically expected value of $MSE_{\hat{Y}}$ for $M = 1, 2, ..., 10$ and plot this as function of M
- What does the situation $M \to \infty$ correspond to?

21) Compare – <u>for the same filter length</u> – the estimated $MSE_{\hat{Y}}$ for the causal FIR Wiener-filter and the non-causal FIR Wiener-filter respectively.

Which is best (for the same filter length)? Why?

F. Non-causal IIR Wiener filter: design, filtering, analysis, performance

Now, estimate the signal $Y(n)$ from the observed signal $X(n)$ when filtering $X(n)$ with a (theoretically) non-causal IRR Wiener-filter defined by

$$H_{wiener}(z) = \sum_{m=-\infty}^{\infty} h(m) \cdot z^{-m}$$

As demonstrated in the figure on page 1, the Wiener-filtered signal $\hat{Y}(n)$ is used as an optimal estimate of $Y(n)$.

22) Set up the Wiener-Hopf equations and find the transfer function $H_{wiener}(z)$ for the optimal non-causal IIR Wiener-filter

23) Find the poles for $H_{wiener}(z)$
Can the filter be implemented both steadily and causally?

24) Find the filter coefficients $\{h(m), \quad -\infty < m < \infty\}$ for the non-causal, stable IIR-filter

25) Calculate the expected estimation error by means of using $\hat{Y}(n)$ for estimation of $Y(n)$ when applying the calculated optimal non-causal IRR-filter, i.e. calculate

$$MSE_{\hat{Y}} = E\left[\left(Y(n) - \hat{Y}(n)\right)^2\right]$$

Compare the value with $MSE_X = E\left[\left(Y(n) - X(n)\right)^2\right]$

An indefinite non-causal IIR-filter cannot be implemented in practice.
Instead, the indefinite non-causal impulse response for the filter can be truncated into a definite non-causal impulse response, corresponding to a non-causal FIR-filter.

26) Calculate the expected $MSE_{\hat{Y}}$ if the calculated filter is truncated to a non-causal filter with 3 filter coefficients.

27) MATLAB-realisation (truncation to filter length = 3):
- Generate $\hat{Y}(n)$ at filtering of $X(n)$ with the truncated non-causal IIR-filter (remember that the filtering must be performed non-causally!!!)
- Plot $\hat{Y}(n)$ and its autocorrelation function and PSD
- Calculate the estimation error $MSE_{\hat{Y}}$ from the concrete signals $Y(n)$ and $\hat{Y}(n)$
- Use, i.e., the commands ***filter, mean, plot/stem, xcorr('biased'), freqz***
- Compare with the theoretical results and comment.

28) Compare for the same filter length = 3 the expected (and realized) $MSE_{\hat{Y}}$ for the truncated non-causal IIR Wiener-filter and the non-causal FIR Wiener-filter respectively.
Which is the best (for the same filter length)?
Why?

G. Causal IIR Wiener filter: design, filtering, analysis, performance

Now we want to estimate the signal $Y(n)$ from the observed signal $X(n)$ at filtering of $X(n)$ with a causal IIR Wiener-filter, defined by

$$H_{wiener}(z) = \sum_{m=0}^{\infty} h(m) \cdot z^{-m}$$

As demonstrated in the figure on page 1, the Wiener-filtered signal $\hat{Y}(n)$ is used as an optimal estimate of $Y(n)$.

As described in the literature, the optimal causal IIR Wiener-filter can be determined as a cascade of a whitening-filter and a MSE-filter. This procedure is used in the following.

29) Write up $S_{xx}(z)$ with the given signal model.

30) Fact $S_{xx}(z)$ so that

$S_{xx}(z) = G^{+}(z) \cdot G^{-}(z)$ where

$G^{+}(z)$ is the min. phase, (all poles and zero-points within the unit circle)

$G^{-}(z)$ is the max. phase, (all poles and zero-points outside the unit circle), and where

$\left|G^{+}(1)\right|^2 = \left|G^{-}(1)\right|^2 = S_{xx}(1)$ (the same DC-amplification)

31) Now determine the stable and causal whitening filter as:

$$G_{whitening}(z) = \frac{1}{G^{+}(z)}$$

32) Then determine the following MSE-filter as:

$$H_{MSE}(z) = causal\ \ part\ of\ \left\{G^{*}_{whitening}(z) \cdot S_{xy}(z)\right\} = causal\ \ part\ of\ \left\{\frac{1}{G^{-}(z)} \cdot S_{yy}(z)\right\}$$

33) Finally, determine the total causal IIR Wiener-filter as:

$H_{wiener}(z) = G_{whitening}(z) \cdot H_{MSE}(z)$

and find the impulse response of the filter

34) Calculate the expected estimation error by means of using $\hat{Y}(n)$ for the estimation of $Y(n)$ when using the calculated optimal causal IIR-filter, i.e. calculate

$$MSE_{\hat{Y}} = E\left[\left(Y(n) - \hat{Y}(n)\right)^2\right]$$

Compare the value with $MSE_X = E\left[\left(Y(n) - X(n)\right)^2\right]$

35) MATLAB-realisation:

- Generate $\hat{Y}(n)$ when filtering $X(n)$ with the causal IIR Wiener-filter
- Plot $\hat{Y}(n)$ and its autocorrelation function and PSD
- Calculate the estimation error $MSE_{\hat{Y}}$ from the concrete signals $Y(n)$ and $\hat{Y}(n)$
- Use, e.g., the commands ***filter, mean, plot/stem, xcorr('biased'), freqz***
- Compare with the theoretical results and comment.
- Compare $MSE_{\hat{Y}}$ for the causal IIR filter with $MSE_{\hat{Y}}$ for causal FIR-filters with various lengths.

H. Conclusion regarding performance comparisons

36) Follow up on the drawn sub-conclusions, e.g.

- What is the theoretically best Wiener-filter?
- What is the best filter, if it is allowed to filter finally non-causally?
- What is the best filter, if filtering is to be performed causally?
- What is the best filter, if the filter length has been determined beforehand?

Problem 4.2 (Theoretical + MATLAB)
Deconvolution and denoising with discrete-time FIR Wiener-filter

Motivation

This exercise deals with the – in practice – common appearing situation, where an observed WSS-signal $X(n)$ consists of a wanted WSS-signal $Y(n)$, which is

- the filter (distorted / blurred)
- noise infected

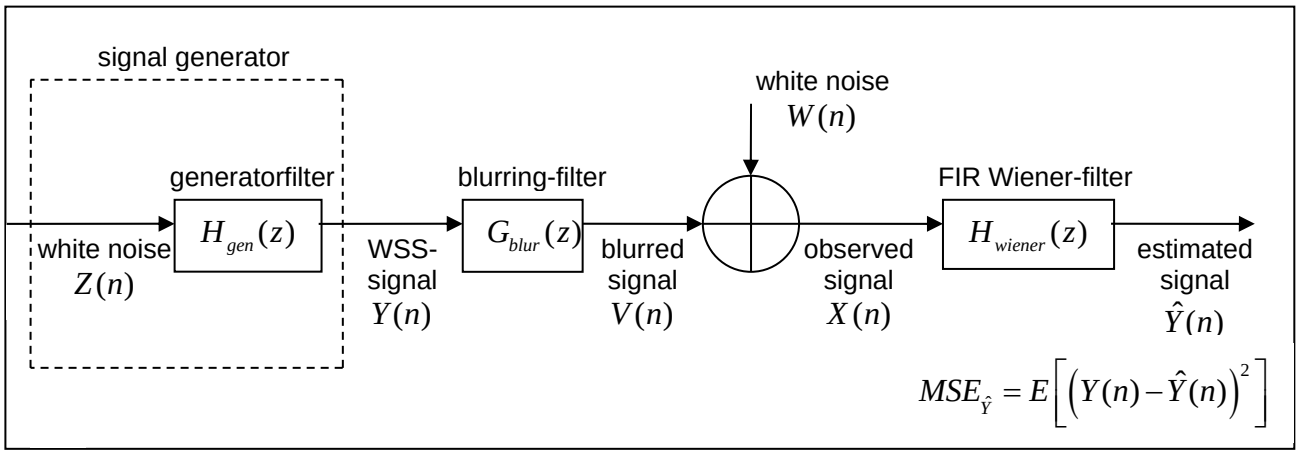
The task of the signal-estimator is therefore to

- compensate for the unwanted filtering (deconvolution / deblurring / inverse filtering)
- reduce the noise (denoising)

There are several deterministic and stochastic methods for combined deconvolution / denoising.
In this exercise we are to study the application of the FIR Wiener filter, i.e. stochastically optimal combined deconvolution / denoising (LMMSEE) of WSS signals, including the performance of the Wiener-filters – both theoretically as well as with simulation in MATLAB.
Regarding the comparisons between the theoretical calculations and simulations, the discrete-time case is dealt with, but all results are absolutely parallel for the continuous-time case.
Besides direct application, the Wiener-filter can be used as reference at comparison of performance for non-optimal deconvolution / denoising methods.

Signal model for the exercise

Below you find a block diagram for producing a WSS signal, which is distorted with a linear filter and is noise infected, and Wiener-filtering of the observed signal with the purpose of estimation of the WSS-signal.



Please note that the signal generator part shown in the figure only is included due to technicalities in the exercise in order to produce a WSS signal $Y(n)$ (and is unchanged from Problem 4.1).

In practical usages, only the observed signal $X(n)$ will exist, and known or estimated knowledge about the stochastic characteristics for the signal $Y(n)$ and the additive noise $W(n)$, and known or estimated knowledge about the blurring filter $G_{blur}(z)$.

Structure of the exercise

The exercise follows the following parts:

A. Theoretical guiding results for determination of optimal filter coefficients.
B. Generation and analysis of the WSS-signal $Y(n)$
C. Definition of and filtering with the blurring-filter
D. Generation and analysis of the WSS-noise $W(n)$
E. Generation and analysis of the observed signal $X(n)$
F. **Causal FIR** Wiener filter: design, filtering, analysis, performance
G. **Non-causal FIR** Wiener filter: design, filtering, analysis, performance
H. Conclusion regarding performance comparisons

For all parts you are to make calculations both theoretically and with MATLAB

A. Theoretical guiding results for determination of optimal filter coefficients.

The performance-function for the Wiener-filter is just like in the standard situation.

$$MSE_{\hat{Y}} = E\left[\left(Y(n)-\hat{Y}(n)\right)^2\right] \quad \text{where}$$

$$\hat{Y}(n) = \sum_{m=-M_1}^{M_2} h(m)X(n-m) \quad \text{i.e. the FIR-causal /non-causal filtering of } X(n)$$

When minimizing $MSE_{\hat{Y}}$ - Set partial derivatives with respect to filter coefficients equal to zero and solve, or equivalently use the orthogonality principle Analog to the standard situation the Wiener Hopf equations which define the optimal filter coefficients h are found.

$$\Sigma_{XY} = \Sigma_{XX} \bullet h \quad \text{where, as usual}$$

$$\Sigma_{XY} = \begin{bmatrix} R_{xy}(-M_1) \\ \dots \\ R_{xy}(M_2) \end{bmatrix} \quad \Sigma_{XX} = \begin{bmatrix} R_{xx}(0) & \dots & R_{xx}(M_1+M_2) \\ \dots & \dots & \\ R_{xx}(M_1+M_2) & & R_{xx}(0) \end{bmatrix} \quad h = \begin{bmatrix} h(-M_1) \\ \dots \\ h(M_2) \end{bmatrix}$$

This means that the only thing missing in order to be able to solve the Wiener-Hopf equations is finding the matrices Σ_{XY} and Σ_{XY} by means of finding the correlation functions $R_{xy}(k)$ and $R_{xx}(k)$ in the concrete signal model for the exercise.

1) Demonstrate, as it is assumed that $Y(n)$ and $W(n)$ are uncorrelated, that
$R_{xy}(k) = g(-k) * R_{yy}(k)$ where $g(k)$ is impulse response for the blurring filter, and demonstrate
$S_{XY}(z) = G^*_{blur}(z) \cdot S_{YY}(z)$
2) Hereafter, demonstrate
$S_{XX}(z) = G_{blur}(z) \cdot G^*_{blur}(z) \cdot S_{YY}(z) + S_{WW}(z)$ whereafter
$R_{xx}(k) = Z^{-1}\{S_{XX}(z)\}$

B. Generation and analysis of the WSS-signal $Y(n)$

(Please note, B is identical with the same part in Problem (13+14).1 so that both the theoretical results as MATLAB-code can be copied from this exercise).

The stochastic WSS signal $Y(n)$ is given by

$R_{yy}(k) = h^{|k|}, \quad h = 0.8$ (autocorrelation)

$S_{yy}(z) = Z\{R_{yy}(k)\} = \frac{1-h^2}{(1+h^2)-h(z+z^{-1})} = \frac{1-h^2}{(1-hz^{-1})(1-hz)}$ (PSD in the z-domain)

$S_{yy}(\omega) = S_{yy}(z)\Big|_{z=e^{j\omega T_s}} = \frac{1-h^2}{(1+h^2)-2h\cos\omega T_s}, \quad f_s = 1/T_s = 10kHz$ (PSD in the frequency domain)

3) Outline $R_{yy}(k)$ og $S_{yy}(\omega)$

4) Show that $Y(n)$ can be found through the differential equation
$Y(n) = h \cdot Y(n-1) + Z(n)$
where $Z(n)$ ("driving noise") is white, additive, zero-mean gaussic noise
(help: calculate $R_{yy}(k)$ from the differential equation)

5) Find the required noise effect of driving noise $\sigma_Z{}^2 = E\left[Z^2(n)\right]$ which ensures that $R_{yy}(0) = 1$ as required in the specification for $Y(n)$ mentioned above.

6) State the transfer function $H_{gen}(z) = \frac{Y(z)}{Z(z)}$ for the signal generator-filter.

7) MATLAB-realisation:
- Generate N=1000 samples of driving noise $Z(n)$ (remember to apply the calculated $\sigma_Z{}^2$)
- Filter $Z(n)$ with signal generator-filter $H_{gen}(z)$
- Plot the resulting WSS signal $Y(n)$ and its autocorrelation function and PSD
- Use, e.g., the commands ***randn, filter, plot/stem, xcorr('biased'), freqz***
- Compare with the theoretical results and comment.

C. Definition of and filtering with the blurring-filter

The linearly distorted blurring filter is given by

$$G_{blur}(z) = \frac{1}{1 - bz^{-1}}, \quad b = 0.5$$

8) MATLAB-realisation:

- Generate $V(n)$ when filtering $Y(n)$ with the specified blurring filter $G_{blur}(z)$
- Plot $V(n)$ and it autocorrelation function and PSD.
- Use, e.g., the commands ***filter, plot/stem, xcorr('biased'), freqz***

D. Generation and analysis of the WSS-noise $W(n)$

(Please note, D is identical with the same part in Problem (13+14).1 so that both the theoretical results as MATLAB-code can be copied from this exercise).

The additive, white, zero-mean, gaussic noise $W(n)$ is given by

$$W(n) = N(0, \sigma_W{}^2), \quad \sigma_W{}^2 = 4$$

$$S_{ww}(z) = \sigma_W{}^2$$

Furthermore the noise $W(n)$ is uncorrelated with the signal $Y(n)$ $(R_{yw}(k) = 0, \quad \forall k \in Z)$

9) Outline $R_{ww}(k)$ og $S_{ww}(\omega)$

10) MATLAB-realisation:

- Generate N=1000 samples of the noise $W(n)$ (remember to use $\sigma_W{}^2$)
- Plot the noise $W(n)$ and its autocorrelation function and PSD.
- Use, e.g., the commands ***randn, plot/stem, xcorr('biased'), freqz***
- Compare with the theoretical results and comments.

E. Generation and analysis of the observed signal $X(n)$

(Please note, E is identical with the same part in Problem (13+14).1 so that both the theoretical results as MATLAB-code can be copied from this exercise).
(Remember however, that the noise is to be added to the blurred signal and not as in the first exercise directly to the wanted signal.)

The observed (noise infected) signal $X(n)$ is given by

$$X(n) = V(n) + W(n), \quad V(n) = g(n) * Y(n)$$

11) MATLAB-realisation:

- Generate the observed signal $X(n)$
- Plot $X(n)$ and its autocorrelation function and PSD.
- Calculate the estimation error MSE_X
- Use, e.g., the commands ***mean, plot/stem, xcorr('biased'), freqz***
- Compare with the theoretical results and comment.

F. Causal FIR Wiener filter: design, filtering, analysis, performance

Now we want to estimate the signal $Y(n)$ from the observed signal $X(n)$ when filtering $X(n)$ with a causal FIR Wiener-filter, defined by

$$H_{wiener}(z) = \sum_{m=0}^{M} h(m) \cdot z^{-m}$$

For the manual calculations $M = 1$ is used, i.e. the sought FIR-filter has two coefficients $h(0)$ and $h(1)$.

12) Write, by means of the results from question 1), an expression for $S_{XY}(z)$

13) Find an expression for $R_{xy}(k), k \in Z$ by means of first partial fraction decomposing $S_{XY}(z)$ and thereafter performing inverse Z-transformation.

(Help: use, e.g., the Z-transformation pairs $a^k \cdot u(k) \xrightarrow[z^{-1}]{z} \frac{1}{1-az^{-1}}, \quad |a| < 1$

$\left(\frac{1}{a}\right)^k \cdot u(-k) \xrightarrow[z^{-1}]{z} \frac{1}{1-az}, \quad |a| < 1$)

14) Write, by means of using the result from question 2) an expression for $S_{XX}(z)$

15) Find an expression for $R_{xx}(k), k \in Z$ by means of first partial fraction decomposing $S_{XX}(z)$ and thereafter perform inverse Z-transformation.

(Help: use, e.g., the Z-transformation pair $a^{|k|} \xrightarrow[z^{-1}]{z} \frac{1-a^2}{(1-az^{-1})(1-az)}, \quad |a| < 1$)

16) Set up the Wiener-Hopf equations as described in part A for $M = 1$ corresponding to $M_1 = 0$ and $M_2 = M = 1$ find the optimal filter coefficients $h(0)$ and $h(1)$ for the causal FIR Wiener-filter.

17) Calculate the expected estimation error by means of using $\hat{Y}(n)$ for estimation of $Y(n)$ when applying the calculated optimal FIR-filter, i.e. calculate

$$MSE_{\hat{Y}} = E\left[\left(Y(n) - \hat{Y}(n)\right)^2\right]$$

18) MATLAB-realisation ($M = 1$):

- Generate $\hat{Y}(n)$ when filtering $X(n)$ with the calculated FIR-filter.
- Plot $\hat{Y}(n)$ and its autocorrelation function and PSD.
- Calculate the estimation error $MSE_{\hat{Y}}$ from the concrete signals $Y(n)$ and $\hat{Y}(n)$
- Use, e.g., the commands ***filter, mean, plot/stem, xcorr('biased'), freqz***
- Compare with the theoretical results and comment.

19) MATLAB-calculation ($M > 1$):
- Set up, in the general case $M > 1$ the Wiener-Hopf equations for determination of the filter coefficients $\{h(m), \quad m = 0,1,...,M-1\}$ for the actual signal model.
- Implement this in MATLAB so that the filter coefficients can be calculated for a random value of M
- Calculate the theoretically expected value of $MSE_{\hat{Y}}$ for $M = 1,2,...,10$ and plot this as the function of M
- What does the situation $M \to \infty$ correspond to?

G. Non-causal FIR Wiener filter: design, filtering, analysis, performance

Now we want to estimate the signal $Y(n)$ from the observed signal $X(n)$ when filtering $X(n)$ with a non-causal FIR Wiener-filter, defined by

$$H_{wiener}(z) = \sum_{m=-M}^{M} h(m) \cdot z^{-m}$$

For the manual calculations, use $M = 1$, i.e. the sought FIR-filter has three coefficients $h(-1)$, $h(0)$ and $h(1)$.

20) Set up the Wiener-Hopf equations as described in part A for $M = 1$ corresponding to $M_1 = M_2 = M = 1$ and find the optimal filter coefficients $h(-1)$, $h(0)$ and $h(1)$ for the non-causal FIR Wiener-filter.
Does the FIR-filter have a linear phase? Why (not)?

21) Calculate the expected estimation error by means of using $\hat{Y}(n)$ for estimation of $Y(n)$ when using the calculated optimal FIR-filter, i.e. calculate

$$MSE_{\hat{Y}} = E\left[\left(Y(n) - \hat{Y}(n)\right)^2\right]$$

22) MATLAB-realisation ($M = 1$):
- Generate $\hat{Y}(n)$ when filtering $X(n)$ with the calculated FIR-filter (remember that the filtering must be performed non-causally!!!)
- Plot $\hat{Y}(n)$ and its autocorrelation function and PSD
- Calculate the estimation error $MSE_{\hat{Y}}$ from the concrete signals $Y(n)$ and $\hat{Y}(n)$
- Use, e.g., the commands ***filter, mean, plot/stem, xcorr('biased'), freqz***
- Compare with the theoretical results and comment.

23) MATLAB-calculation ($M > 1$):

- Set up, in the general case, $M > 1$ the Wiener-Hopf equations for determination of the filter coefficients $\{h(m), \quad m = -M, ..., -1, 0, 1, ..., M\}$ for the actual signal model.
- Implement this in MATLAB so that the filter coefficients can be calculated for at random value of M
- Calculate the theoretically expected value of $MSE_{\hat{Y}}$ for $M = 1, 2, ..., 10$ and plot this as a function of M
- What does the situation $M \to \infty$ correspond to?

H. Conclusion regarding performance-comparisons

24) Compare for the same length of filter the expected $MSE_{\hat{Y}}$ for the causal FIR- Wiener-filter and the non-causal FIR Wiener-filter respectively.
Which is the best (for the same length of filter)?
Why?